Data Diggers: A Multiview Framework for Fraudulent Review Detection using Deep Embeddings and Stylometric Clustering

Felicia Huffman
CS5831 - Advanced Data Mining
Michigan Technological University
Houghton, Michigan
fmhuffma@mtu.edu

Elisabeth MacChesney
CS5831 - Advanced Data Mining
Michigan Technological University
Houghton, Michigan
emacches@mtu.edu

Jessica Pamela Feliz Garrido
CS5831 - Advanced Data Mining
Michigan Technological University
Houghton, Michigan
jfelizga@mtu.edu

Riley Meeves
CS5831 - Advanced Data Mining
Michigan Technological University
Houghton, Michigan
rjmeeves@mtu.edu

***Abstract*—The proliferation of generative artificial intelligence has led to a significant increase in fraudulent and AI-generated reviews across major e-commerce platforms. This project proposes a hybrid detection framework that isolates fraudulent reviews using a combination of behavioral feature engineering, deep semantic embeddings, and multiview clustering. Supervised baseline models were evaluated to establish a performance benchmark, with Logistic Regression achieving a leading F_1-score of 0.917 and an accuracy of 85.10%. To capture deeper contextual and structural anomalies, Convolutional Neural Networks (CNN) and DeBERTa embeddings were utilized, and their separability was visualized using Uniform Manifold Approximation and Projection (UMAP). The integration of stylometric analysis with behavioral metadata provides a foundation for real-time anomaly detection in highly imbalanced e-commerce datasets.**

***Index Terms*—fraud detection, anomaly detection, deep learning, stylometry, natural language processing, clustering**

I. Introduction

Understanding the veracity of online reviews is important for consumers and e-commerce platforms alike. Due to the increasing sophistication of large language models, malicious actors can generate highly convincing fake reviews at scale, skewing product ratings and deceiving consumers. Traditional detection methods, which rely heavily on simple keyword matching or basic account metadata, are increasingly insufficient against these nuanced attacks. Previous studies have explored rating deviations [2] and graph neural networks [3] to combat this, but adversarial adaptations continue to outpace traditional models.

This study explores a comprehensive, multi-view methodology for detecting fraudulent reviews. By combining behavioral metadata (e.g., posting frequency, rating deviation) with deep semantic embeddings derived from Natural Language Processing (NLP) models, we aim to capture both the hidden patterns of user behavior and the linguistic signatures of synthetic text.

The primary contributions of this paper include:

1) The extraction and evaluation of 14 distinct behavioral features mapped against class distributions.
2) A comparative analysis of eight supervised baseline models, optimized using Optuna.
3) The application of DeBERTa and CNN embeddings to extract deep semantic and structural representations of review text, visualized via UMAP.

II. Related Work

The detection of fraudulent reviews has evolved significantly as malicious actors adapt to platform defenses. Early approaches primarily focused on analyzing behavioral metadata and rating patterns. For instance, Savage et al. [2] demonstrated the effectiveness of identifying opinion spam through anomalous rating deviations. As adversarial tactics grew more sophisticated, researchers began modeling complex reviewer-product networks, with Chen et al. [3] employing graph neural networks to detect organized astroturfing campaigns.

More recently, deep learning and natural language processing have become central to identifying synthetic and spam text. Mewada and Dewang [1] provide a comprehensive overview of state-of-the-art detection methods, highlighting the necessary shift toward semantic analysis in the face of advanced language models.

While these studies present robust solutions within their specific domains, they often isolate behavioral analysis from linguistic analysis. Our project builds upon these foundational works by proposing a multiview framework that explicitly bridges this gap, integrating the structural context provided

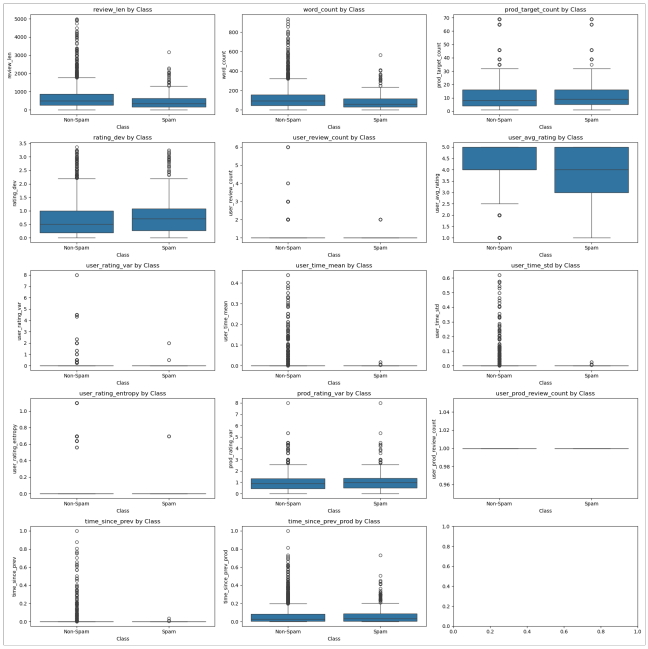

Fig. 1. Distribution of engineered behavioral features mapped against fraudulent and legitimate review classifications.

by deep embeddings (DeBERTa and CNNs) with traditional behavioral clustering.

III. Methodology

A. Feature Engineering

To capture anomalous user behavior, 14 distinct features were engineered from the raw review data. These features span temporal inconsistencies, rating variations, and basic stylometric lengths. Key engineered features, whose distributions across classes are visualized in Fig. 1, include:

- **Temporal Features:** *time_since_prev* (normalized time since the user's last review), *time_since_prev_prod*, *user_time_mean*, and *user_time_std*.
- **Rating Variations:** *user_rating_var* (inconsistency in user ratings), *user_avg_rating*, *user_rating_entropy*, *prod_rating_var*, and *rating_dev*.
- **Volume and Length Metrics:** *user_review_count*, *user_prod_review_count*, *prod_target_count*, *review_len* (character length), and *word_count*.

B. Deep Embeddings

Relying solely on metadata leaves the system vulnerable to attackers who mimic normal behavioral cadence. Therefore, text-based embeddings were generated to analyze the semantic and structural content of the reviews. A pre-trained DeBERTa (Decoding-enhanced BERT with disentangled attention) model was utilized to extract rich semantic representations, capturing deep contextual relationships within the text. Concurrently, a Convolutional Neural Network (CNN) architecture was employed to extract localized, n-gram-like structural patterns.

C. Multiview Spectral Clustering

To fuse the distinct perspectives of behavior and linguistics, a multiview framework was designed. By projecting the high-dimensional embeddings into a lower-dimensional space using UMAP, distinct clusters of anomalous behavior were identified. This approach allows for the unsupervised detection of novel spam campaigns that lack historical training labels.

IV. Experimental Setup

A. Data Description

The dataset utilized for this study is a 10% stratified sample consisting of 35,896 total user reviews. An initial exploratory data analysis revealed a severe class imbalance, heavily skewing towards legitimate, non-spam reviews (32,210 reviews) compared to fraudulent, spam reviews (3,686 reviews), as illustrated in Fig. 2. This imbalance necessitated the use of appropriate evaluation metrics, prioritizing the F_1-score and Receiver Operating Characteristic - Area Under the Curve (ROC-AUC) over standard accuracy.

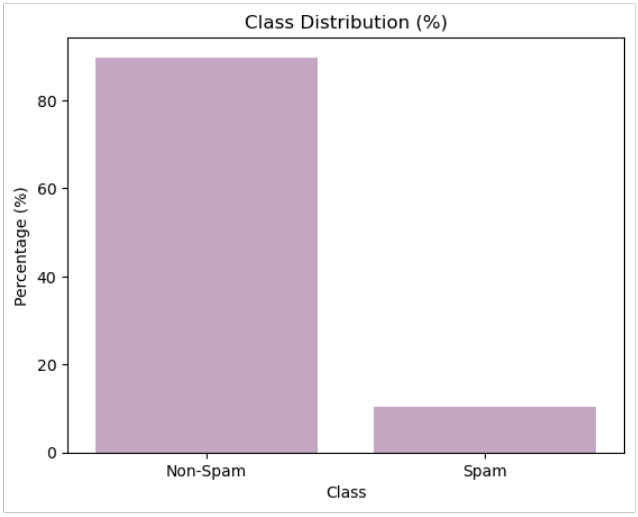

Fig. 2. Class distribution of the dataset sample, highlighting the severe imbalance between legitimate (non-spam) and fraudulent (spam) reviews.

B. Model Training and Optimization

Eight supervised learning models were established as baselines: Support Vector Machine (SVM), Neural Network (Multi-layer Perceptron), Logistic Regression, XGBoost, Decision Tree, Random Forest, Naive Bayes, and a Stacking Classifier. Hyperparameter tuning was conducted using Optuna, a define-by-run framework that employs Tree-structured Parzen Estimator (TPE) algorithms to efficiently navigate the search space. All models were trained and evaluated locally.

V. Results

A. Supervised Baseline Performance

The models were evaluated based on their ability to correctly identify the minority fraudulent class while minimizing

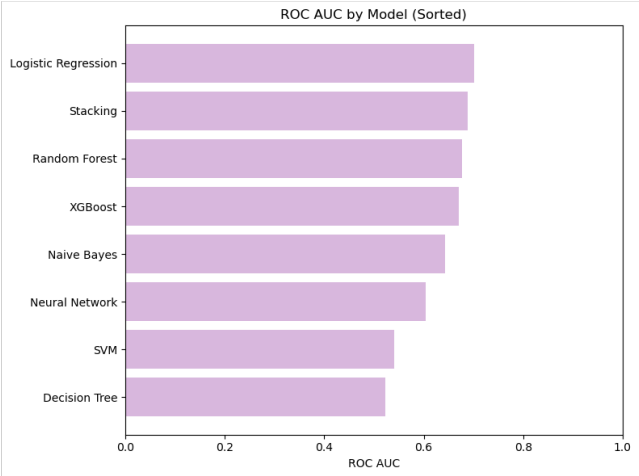

Fig. 3. Comparison of the Receiver Operating Characteristic (ROC) Area Under the Curve (AUC) across all evaluated baseline models.

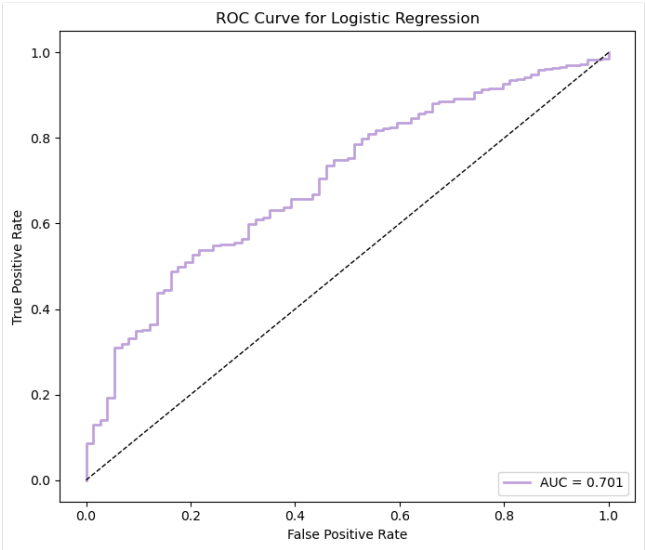

Fig. 4. Receiver Operating Characteristic (ROC) curve for the top-performing Logistic Regression model.

false positives. The results demonstrate a significant variance in model efficacy, detailed visually in Fig. 3 and quantitatively in Table I.

TABLE I
PERFORMANCE METRICS OF SUPERVISED BASELINE MODELS

Model	**ROC-AUC**	F_1**-Score**	**Accuracy (%)**
Logistic Regression	**0.701**	**0.918**	**85.10**
Random Forest	0.677	0.722	59.61
Neural Network	0.604	0.708	57.66
XGBoost	0.670	0.647	51.95
Stacking Classifier	0.689	0.608	48.61
Decision Tree	0.523	0.651	51.25
Naive Bayes	0.643	0.546	43.04
SVM	0.541	0.034	11.84

Logistic Regression outperformed all other baseline configurations, achieving a peak F_1-score of 0.917 and an overall accuracy of 85.10% with an L1 penalty constraint, as shown by Fig. 4. The Neural Network also demonstrated strong predictive capability, yielding an F_1-score of 0.708.

Conversely, the SVM model struggled significantly with the feature space, resulting in an F_1-score of only 0.034 and an accuracy of 11.84%, despite class weight balancing. Ensemble methods showed moderate success; Random Forest achieved an AUC of 0.677 and an F_1-score of 0.722, while the XGBoost model reached an AUC of 0.670 and an F_1-score of 0.647.

B. Embedding Separability

To assess the discriminative power of the unsupervised methods, UMAP projections were generated for the behavioral features, the CNN embeddings, and the DeBERTa embeddings.

The behavioral-only projection (Fig. 5) showed dense overlapping, confirming that behavioral metrics alone struggle to isolate sophisticated spam. However, the DeBERTa and CNN projections (Fig. 6 and Fig. 7) revealed tighter, more distinct

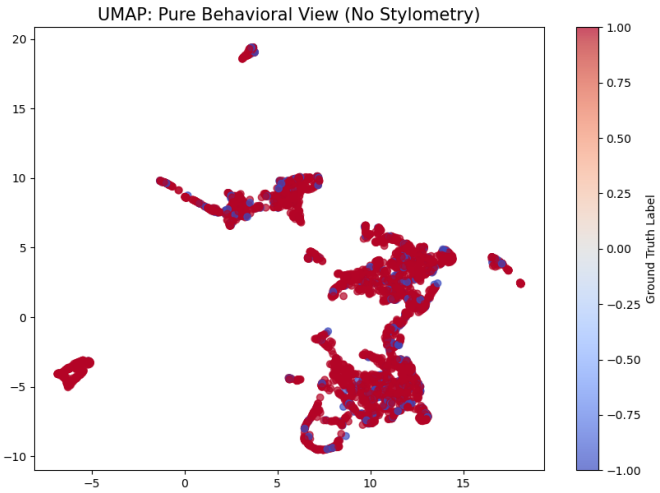

Fig. 5. UMAP projection based purely on behavioral metadata, demonstrating limited distinct separability.

topological structures, indicating that deep linguistic features capture latent anomalies missed by metadata analysis.

VI. DISCUSSION AND CONCLUSION

The experimental results underscore the necessity of a multi-faceted approach to fraudulent review detection, though significant challenges remain. While Logistic Regression proved highly effective on the engineered feature set, its reliance on static variables makes it susceptible to adversarial adaptation.

The integration of DeBERTa and CNN embeddings provided a crucial layer of semantic security, and UMAP visualizations confirmed that these deep models capture subtle stylometric deviations. However, the anticipated unsupervised

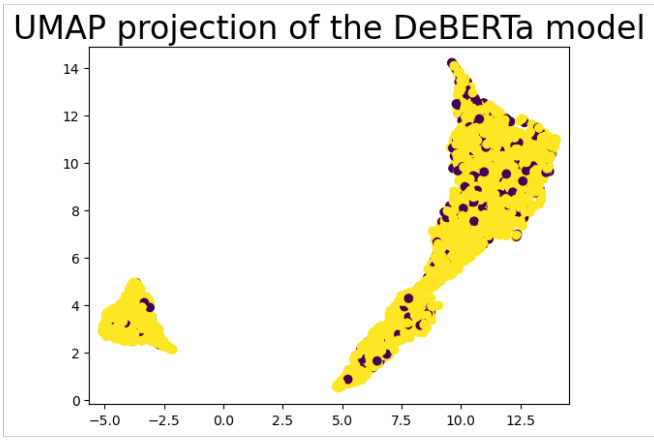

Fig. 6. UMAP projection of DeBERTa semantic embeddings, showing improved structural grouping.

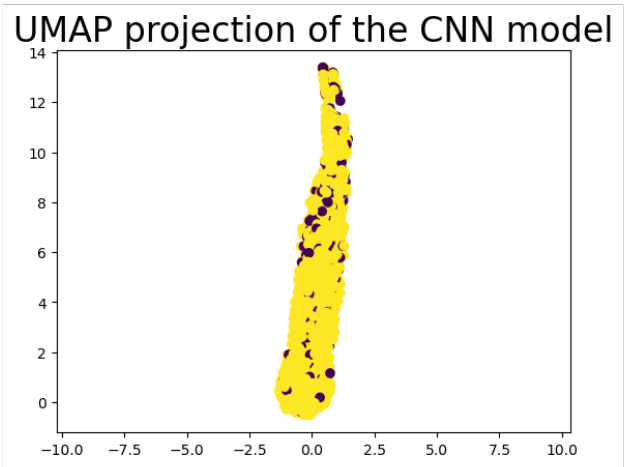

Fig. 7. UMAP projection of CNN structural embeddings, illustrating localized anomaly clustering based on n-gram patterns.

anomaly detection using Spectral Clustering performed poorly in practice.

Several key limitations impacted the final framework:

- **Unsupervised Performance:** Despite visual separability in lower dimensions, the actual multiview spectral clustering algorithms struggled to accurately group anomalies.
- **Class Imbalance:** The severe skew toward legitimate reviews heavily constrained model precision and led to mid-level results across several ensemble baselines.
- **Computational Costs:** Fusing advanced deep learning embeddings required extensive matrix operations that exceeded available local computing resources, limiting the scale of evaluation.

To address these limitations, future work will involve scaling the framework using higher-performance computing clusters to process the full, non-sampled dataset. Methodologically, incorporating cross-validation will improve model generalization, and exploring Isolation Forests may yield more robust unsupervised anomaly detection than spectral methods. Finally, deeply analyzing behavioral feature importance and expanding this framework for real-time inference across other AI-prevalent platforms remains a priority for continuing research.

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SELF-DECLARATION FORM: PROJECT CONTRIBUTIONS

This section outlines the individual contributions of each team member to the successful completion of the "Data Diggers: Fraudulent Review Detection" project. To ensure comprehensive collaboration, each core component was managed by specific members who subsequently assigned and distributed tasks to the rest of the group, ensuring that every team member meaningfully contributed to all aspects of the final deliverables.

- **Felicia Huffman:** Managed the Jupyter Notebook codebase and development environment. Responsible for structuring the analytical pipeline and assigning specific programming, preprocessing, and model implementation tasks to the group so each member could actively contribute to the core code.
- **Elisabeth MacChesney:** Managed the creation and structuring of the final presentation slideshow. Responsible for the visual narrative and delegating slide creation, figure synthesis, and talking-point development tasks to ensure equitable contribution to the presentation.
- **Jessica Pamela Feliz Garrido:** Co-managed the drafting and formatting of the IEEE conference paper alongside Riley. Responsible for synthesizing technical results and delegating section-writing tasks to guarantee all group members meaningfully contributed to the academic report.
- **Riley Meeves:** Co-managed the drafting and formatting of the IEEE conference paper alongside Jessica. Responsible for ensuring strict compliance with IEEE standards, integrating multi-view methodology narratives, and delegating report tasks across the team to ensure comprehensive documentation.